of retirement assets.[59] Retirement asset managers have been able to hold onto higher-fee products partly because fees on 401(k) plans were not required to be disclosed until 2012. As more investors pay attention to fees in retirement plans, money should start flowing to ETFs. The process will be slow, though, because consultants and mutual fund trustees control a lot of these funds, and fund beneficiaries have little say.

Too Much Trading?

There is another way that asset management companies can make money off ETFs, even though the margins on managing ETFs are small. John Bogle, the founder of Vanguard, says:[60] "The trick of ETFs . . . is that the costs of administration are basically thrown over to the marketplace, so people pay for them with their brokerage commissions and things of that nature."

Some asset management companies can subsidize ETFs by making money from trading costs—especially if they have their own trading platforms or work hand-in-hand with brokers. Fidelity Investments, for example, began charging high fees in March 2013 for investors who sell "commission-free" ETFs on its trading platform within short time periods.[61] Many ETF sponsors, however, do not directly benefit from their costs of trading. ETF bid–ask spreads average 0.9%, but fixed income and equity funds have bid–ask spreads below 0.25%.[62] The biggest ETFs, however, have very small trading costs and the ten largest ETFs from 2007 through 2012 have average spreads of 0.04%.[63] Nevertheless, even if you trade the biggest ETFs a lot, the expenses start to add up very quickly—and cost many times more than a traditional (buy and infrequently trade) active mutual fund. Mutual funds, in contrast, have zero bid–ask spreads.[64]

Leveraged ETFs

Retail investors must be particularly careful with leveraged ETFs. With these products, daily compounding causes returns over more than one day to differ from the underlying index. Suppose you invest $100 in a traditional fund and $100 in a hypothetical "3×Leveraged Fund" based on the same index. The index rises 10%.

[59] Computed by Cerulli Associates, as quoted by Weinberg, A. I., What ETFs' Next Act Will Look Like, *Wall Street Journal,* Oct 23, 2012.

[60] John Bogle, testimony to Senate Committee on Government Affairs Hearing on "Mutual Funds: Trading Practices and Abuses that Harm Investors, November 3, 2003.

[61] Grind, K., "Fidelity's ETF Fee Spurs a Backlash," *Wall Street Journal* , Mar 14, 2013.

[62] Numbers from IndexUniverse, quoted by Coombes, A., "Calculating the Costs of an ETF," *Wall Street Journal*, Oct. 23, 2012.

[63] Numbers computed by Paul Tetlock.

[64] In mutual funds, current shareholders absorb the liquidity costs of marginal traders, whereas the liquidity costs are borne by the traders themselves in ETFs. Because of these different liquidity profiles, Guedj and Huang (2009) show that mutual funds and ETFs will co-exist in equilibrium and attract different clienteles with different liquidity needs.